

Madhumitha Juluri

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SUMMARY

Entry-Level Software Engineer with a strong academic background and hands-on internship experience at top tech firms. Proficient in Java, Python, and JavaScript, with solid understanding of object-oriented programming, algorithms, and version control using Git. Successfully developed and tested full-stack applications in agile settings, demonstrating strong problem-solving and collaboration skills. Eager to contribute, learn, and grow under expert mentorship in a remote environment.

EDUCATION

Christian Brothers University Aug 2023 - Dec 2024
Master of Science, Computer Information Systems

- **GPA:** 3.85/4.0
- **Achievements:** Graduate Research Assistant, Magna Cum Laude

Sri Indu College of Engineering & Technology Aug 2018 - May 2022
Bachelor of Technology, Computer Science

- **Achievements:** Summa Cum Laude, Programming Club President, 3-time ACM-ICPC Regional Participant

PROFESSIONAL EXPERIENCE

Comcast Jan 2025 - Present
Sr Software Engineer - Data Platform Engineering Memphis, TN

- Developed and maintained a microservices data platform using Spring Boot, Kafka, and Kubernetes to process 800TB+ daily data from 60M+ subscribers with 99.99% uptime, applying object-oriented programming principles.
- Engineered a real-time analytics engine with Apache Flink and Elasticsearch to deliver millisecond insights for network optimization across 40+ data centers.
- Designed a fault-tolerant streaming infrastructure using Kafka and Redis to handle 15M+ events/second with automatic failover capabilities.
- Implemented an API gateway using Kong with OAuth 2.0/JWT authentication to manage over 500 microservices for both internal and external integrations.
- Led a cross-functional team of 15 developers by establishing coding standards and architecture decisions, significantly boosting agile development practices.
- Optimized distributed system performance with advanced caching strategies and database tuning, achieving an 85% reduction in response times while utilizing version control (Git) for collaborative code management.

Intuit Sep 2024 - Dec 2024
Software Engineer Intern - FinTech Platform Engineering Remote

- Created a financial analytics platform with Spring Boot, React, and PostgreSQL to process 150M+ tax transactions and support real-time fraud detection, emphasizing clean code documentation.
- Transitioned a legacy monolithic application to a microservices architecture using Spring Cloud, Docker, and Kubernetes, which fostered agile collaboration and improved system resilience by 90%.
- Implemented an ML pipeline using Python and TensorFlow for financial risk assessment, reducing false positives by 65% and highlighting a methodical approach to testing and debugging.
- Developed a low-latency payment service with Java, Kafka, and Redis to handle 2M+ daily transactions while ensuring compliance with PCI DSS standards.
- Conducted comprehensive performance testing using JMeter to guarantee system reliability during 30x traffic spikes, supporting robust agile sprint reviews.
- Established a data quality framework with automated validation and lineage tracking, documenting technical processes for critical financial datasets.

Bank of America Jun 2024 - Aug 2024
Software Engineer Intern - Investment Banking Technology Charlotte, NC

- Developed ultra-low-latency trading infrastructure using Java, Spring Boot, and Kafka to support 750K+ concurrent connections with microsecond-level processing, reinforcing principles of object-oriented programming.
- Built a high-performance streaming service with Kafka Streams and Redis to deliver live market data to 30K+ traders with sub-3ms latency, ensuring rapid troubleshooting and system stability.
- Implemented a risk management system using Python, PostgreSQL, and React for real-time portfolio analysis on \$5B+ daily trading volume, with attention to detailed code documentation.
- Architected a multi-tier caching solution using Redis Cluster to optimize data access and reduce database load by 92%, aligning with best practices in agile debugging.
- Designed secure REST and GraphQL APIs with OAuth 2.0 authentication, ensuring SOX compliance and demonstrating strong collaboration on cross-functional projects.
- Established GitOps workflows with ArgoCD and implemented infrastructure as code using Terraform to integrate DevOps practices.

HRI Consultants

Jul 2022 - Jul 2023

Senior Software Engineer - Data Engineering

New Delhi, India

- Architected a scalable data platform using Spring Boot, Apache Spark, and MySQL to process 80TB+ daily data from 25+ clients, ensuring high reliability (99.97%) and incorporating core software development practices.
- Built robust ETL pipelines with Apache Airflow, Python, and Spark, orchestrating complex workflows with automated error handling and thorough code documentation.
- Implemented a high-throughput real-time streaming solution using Apache Kafka and Spark Streaming to enable analytics for financial and healthcare industries.
- Executed a comprehensive AWS cloud migration with auto-scaling and multi-region deployment, achieving a 45% cost reduction and fostering agile team collaboration.
- Mentored a team of 8 engineers, established coding standards, and conducted architectural reviews to improve code quality in a dynamic, agile environment.
- Established a comprehensive data quality framework that reduced issues by 50% and enhanced data lineage visibility, supporting systematic debugging processes.

MyGate

Mar 2021 - May 2022

Backend Developer - PropTech Platform

Hyderabad, India

- Developed a scalable security and community management platform using Node.js, Express.js, and MongoDB, serving 2M+ residents across 15K+ complexes and adhering to best practices in clean code and documentation.
- Engineered a real-time messaging and notification system with WebSocket, Redis, and Apache Kafka to handle 5M+ daily messages with instant delivery, focusing on agile troubleshooting techniques.
- Designed secure payment processing solutions via Razorpay and Stripe APIs for 500K+ residents, incorporating automated billing, reconciliation, and careful system debugging.
- Architected RESTful APIs using Express.js to support 1M+ monthly active users with 99.5% uptime and sub-100ms response times, ensuring maintainability and clear technical documentation.
- Optimized MongoDB performance through indexing and query refinement, reducing response times by 70% and supporting horizontal scaling in an agile development setting.
- Implemented IoT integration for smart apartment features that process sensor data from 50K+ devices, reinforcing robust development and debugging processes.

Zenoti

Jul 2020 - Mar 2021

Software Engineer - SaaS Platform

Hyderabad, India

- Developed a cloud-native spa and salon management platform using Java, Spring Boot, and PostgreSQL, serving 15K+ businesses globally and emphasizing clear code documentation.
- Implemented a scalable multi-tenant SaaS architecture with tenant isolation that supports 100K+ concurrent users at 99.9% uptime, aligning with agile team practices.
- Built an intelligent booking engine using Java and Redis to handle 500K+ monthly appointments with real-time availability, integrating essential troubleshooting methods.
- Developed secure payment infrastructure using Stripe and PayPal APIs to process \$50M+ annual transactions in compliance with PCI standards, ensuring continuous testing and debugging.
- Designed RESTful APIs and third-party integrations for POS systems and accounting software to automate business processes, emphasizing documentation and collaboration.
- Created a business analytics dashboard that provides real-time insights and revenue reporting for spa and salon owners, showcasing effective agile communication and process documentation.

PROJECTS & CERTIFICATIONS

Real-time Data Platform

- Architected cloud-native data platform using Kafka, Flink, and Kubernetes with 8K+ GitHub stars and 500+ production deployments

Financial Analytics Dashboard

- Built comprehensive financial analytics platform using React, Spring Boot, and Snowflake supporting 25K+ concurrent users

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, Scala, SQL, Bash, Go, C++, Kotlin, C#
- **Backend Technologies:** Spring Boot, FastAPI, Node.js, Express.js, Django, Flask, Apache Spark, Kafka Streams, Microservices
- **Big Data & Analytics:** Apache Spark, Hadoop, Kafka, Flink, Elasticsearch, dbt, Airflow, Beam, Trino, Presto
- **Frontend Technologies:** React, Next.js, Vue.js, TypeScript, HTML5, CSS3, Tailwind CSS, Material-UI, D3.js, Angular
- **Databases & Storage:** PostgreSQL, MongoDB, Redis, Snowflake, BigQuery, DynamoDB, Cassandra, Neo4j, MySQL
- **Cloud & Infrastructure:** AWS, Azure, GCP, Docker, Kubernetes, Terraform, CloudFormation, Serverless, Lambda
- **DevOps & CI/CD:** GitHub Actions, Jenkins, Docker, Kubernetes, Helm, ArgoCD, Prometheus, Grafana, Ansible
- **Data Technologies:** Snowflake, Databricks, Apache Airflow, Great Expectations, dbt, Fivetran, Apache Atlas
- **Testing & Quality:** Jest, PyTest, JUnit, Selenium, Cypress, SonarQube, ESLint, TestNG, Mockito
- **Machine Learning:** TensorFlow, PyTorch, Scikit-learn, MLflow, Kubeflow, Apache Spark ML, H2O.ai
- **Core Computer Science:** Object-oriented Programming, Basic Algorithms/data Structures, Version Control (Git)

CERTIFICATION

- AWS Solutions Architect Professional
- Google Professional Data Engineer
- Azure Data Engineer Associate